

Android emulator

Article • 01/25/2024

The Android emulator, which is produced by Google, simulates Android devices on your computer so that you can test your apps on a variety of devices and Android API levels, without needing to have each physical device. When you deploy and test your app on the emulator you select a pre-configured or custom virtual device, known as an Android Virtual Device (AVD) that simulates a physical Android device such as a Pixel phone. Alternatively, you can deploy your app to a physical device. For more information about deploying your app to physical device, see [Set up Android device for debugging](#).

The Android emulator is installed by Visual Studio, and an AVD created, when you build your first .NET Multi-platform App UI (.NET MAUI) app for Android. For more information, see [Build your first app](#).

If hardware acceleration isn't available or enabled, the emulator will run very slowly. You can significantly improve the performance of the emulator by enabling hardware acceleration and by using virtual device images that are appropriate for your processor architecture. For more information, see [How to enable hardware acceleration with Android emulators](#).

Physical Android devices are emulated by AVDs, which specify the Android version and hardware characteristics of the simulated device. To test an app, create AVDs that model each device your app is designed to run on. Each AVD runs as an independent device with its own storage for user data, and SD card data. The emulator stores the user data, and SD card data in a folder specific to the AVD. When you launch the emulator, it loads the user data and SD card data from the AVD folder. For information about managing virtual devices, see [Managing virtual devices with the Android Device Manager](#). For information about editing the configuration of a virtual device, see [Editing Android virtual device properties](#).

After you've created an AVD you can launch the emulator and debug an app in it. For more information, see [Debug on the Android emulator](#). Once an app is installed on an AVD, you can run it from the device as you would run any app on a device.

In addition, the emulator provides versatile networking capabilities that can be used for different purposes, including connecting to an emulator running on a Mac from inside a Windows virtual machine (VM). For more information, see [Connect to an Android emulator on a Mac from inside a Windows virtual machine](#).

For information about diagnosing emulator issues, see [Android emulator troubleshooting](#).

Use the emulator

When the Android emulator is running, you use your computer mouse pointer to mimic your finger on the touch screen, select menu items and input fields, and click buttons and controls. You can type in the emulator by using your computer keyboard or by using the onscreen keyboard that appears in the emulator.

Common actions can be performed in the emulator via its panel on the right hand side:

 Expand table

Action	Icon	Description
Close		Close the emulator.
Minimize		Minimize the emulator window.
Power		Click to turn the screen on/off. Click and hold to turn the device on/off.
Volume up		Increase the volume.
Volume down		Decrease the volume.
Rotate left		Rotate the screen 90 degrees left.
Rotate right		Rotate the screen 90 degrees right.
Take screenshot		Click to take a screenshot of the device.
Enter zoom mode		Click to change the cursor to the zoom icon. Click again to exit zoom mode.
Back		Return to the previous screen, or close an options menu, dialog, onscreen keyboard, or Notifications panel.
Home		Return to the Home screen.
Overview (recent apps)		Click to open a list of apps you've worked with recently. To open an app, tap its thumbnail.
More		Click for more features and settings.

The following gestures can be simulated in the emulator:

 Expand table

Gesture	Description
Tap	Press and release the primary mouse button.
Double tap	Double-click and then release the primary mouse button.
Drag	Press and hold the primary mouse button, move the item, and then release.
Touch and hold	Press the primary mouse button, hold, and release.
Pinch	Press the CTRL key on Windows, or the Command key on macOS, to display a pinch gesture multi-touch interface. The mouse acts as the first finger, which can be moved by dragging the cursor. The second finger is across from the anchor point. Pressing the left mouse button simulates touching both points, and releasing simulates removing both points.
Swipe	Press and hold the primary mouse button, swipe, and then release.
Vertical swipe	Open a vertical menu on the screen and use the mouse scroll wheel to scroll through the menu items.

Advanced emulator scenarios

You might want to test your app on a virtual device using more than just basic touch gestures. For example, you might want to simulate different network conditions. The following table lists where to find more information about advanced Android emulator scenarios:

 Expand table

Scenarios	More information
Use the camera	Camera support  on developer.android.com
Install and add files	Install and add files  on developer.android.com
Use Wi-Fi	Wi-Fi  on developer.android.com
Use extended controls	Extended controls, settings, and help  on developer.android.com
Use snapshots	Snapshots  on developer.android.com

Scenarios	More information
Use the emulator from the command line	Start the emulator from the command line  on developer.android.com
Send console commands	Send emulator console commands  on developer.android.com
Set up networking	Set up Android Emulator networking  on developer.android.com
Configure hardware acceleration	Configure hardware acceleration for the Android Emulator  on developer.android.com
Android emulator tools	Comparison of Android Emulator tools  on developer.android.com

Limitations

The Android Emulator doesn't include virtual hardware for the following:

- Bluetooth
- NFC
- SD card insert/eject
- Device-attached headphones
- USB